

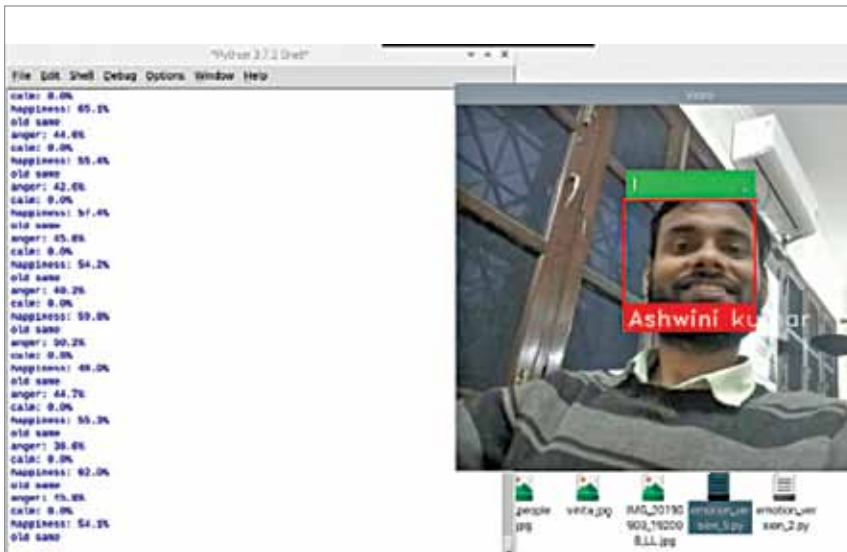


Figure 16: Face detection

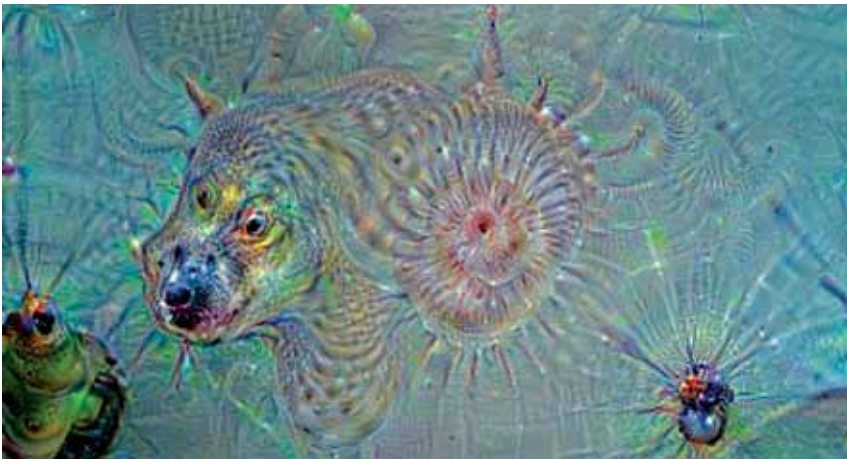


Figure 17: Random robot dream



Figure 18: Robot's dream generated from the saved incident captured by camera



Figure 19: Original picture



Figure 20: The robot's dream generated from the picture in Figure 19

git clone <https://github.com/JoBergs/PsyCam>

Then open the PsyCam folder and change the file path for dream creating image in *utilities.py* to the same path where the captured frame of *facrecog.py* image was saved. In the same folder, create a new Python file and then a script that runs the *testpsycam.py* file in certain intervals of time and makes it go to sleep mode, that is, off/on.

To see the dream image created by the robot, open the dream folder that is situated in the PsyCam folder. This completes the coding.

Connection and testing

Connect the Raspberry Pi Type-C USB port with 5V cable. Next, connect the Raspberry Pi camera using a ribbon cable to the Raspberry Pi camera port. Connect the eyes OLEDs and other parts as described in the earlier AI robot article published in November 2020 issue.

After setting and saving all the codes, test various features of the robot. Run the *face.py* and *run.py* scripts, and wait for a few seconds. The screen will then start showing the face and the robot will start talking according to the face it detects (see Figure 16).

To check the dream of the robot, run the code *pycam.py*. The robot will start capturing the pics and start creating a dream. You may open the dreams folder and see the saved picture of the robot's dream (see Figure 17 through Figure 20).

Note: The source code of this project is available for free download at source.efymag.com

END 

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